

A Parametric 3D Knitting Workflow for Punchcard Knitting Machines

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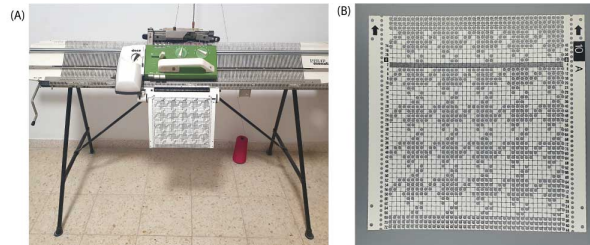


Figure 1: A Passap duo 80 knitting machine (A). Standard commercially available punchcard (B).

ABSTRACT

Punchcard knitting machines allow automatic needle selection based on a punched pattern. The machine reads the card and translates a punched hole to specific knitting actions. Punchcards are usually purchased ready to use or manually punched by the user. Current tools allow the digital fabrication of punchcards, however those tools focus on two-dimensional graphic patterns. This work introduces a parametric workflow for designing and fabricating custom-made punchcards that enable 3D knitting and shaping. We demonstrate our workflow using two case studies that implement short-row knitting to obtain 3D surfaces and non-rectangular shapes. In the first, we generate parametric spike 3D textures, and in the second, we demonstrate long curved shapes and 3D cones. A DXF file is saved and used for cutting the punchcard using a digital desktop cutting machine. Our workflow broadens the design opportunities when using punchcard-driven domestic knitting machines.

CCS CONCEPTS

• **Computing methodologies** → Computer graphics; Shape modeling; Parametric curve and surface models; • **General and reference** → Cross-computing tools and techniques; Design.

KEYWORDS

Parametric Design, Partial knitting, Shaping, Punchcards

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1 INTRODUCTION

Domestic flatbed knitting machines allow relatively fast fabrication of knitted textiles compared to hand knitting. These machines are used by a wide range of knitters to fabricate personalized knitted objects, such as hobbyists, makers, small business owners, and researchers. Domestic flatbed knitting machines are constructed of either single or double needle beds and a knitting carriage. The knitting operation involves moving the carriage from one side of the beds to the other, which lifts needles and forms loops around them. Each repeating stroke of the carriage can form a loop within the one that was formed below it, creating the knitted fabric [1]. We have identified three main types of domestic knitting machines that illustrate the technological advances in the needle selection process: the first operates by pushing buttons, punch cards drive the second, and the third is electronic.

The knitting process of all three machines is based on the same logic: selecting operating needles and determining the knitting action. Generally, each needle can knit, tuck, or not operate. The sequence and the action being done forms the knitted structure. While the knitting action is being set on the carriage, the needles are actuated differently on each type of machine. The Brother's kh-588 pushbutton knitting machine, for example, uses eight-button combinations to determine which needle is set in the working position. The sequence may repeat itself when more than eight needles are used. To change the knitting sequence, the user manually presses the buttons after completing the required set of rows [2].

Punchcard knitting machines operate similarly but allow automatic needle selection based on the punched pattern. A punched hole indicates a certain knitting operation, while an unpunched area indicates a different one. The selection of which needle actions is used is made on the carriage. Each machine brand uses a specific card proportion that controls a different range of needles. Like pushbutton machines, the punchcard's pattern repeats when the number of needles used exceeds the range specified by the card.

Traditionally cards could be purchased ready to use from the machine's manufacturer or manually punched based on a pre-made design using special punching tools. Knitters may create their own design; hand punch their own punchcards or manually alternate working needles to knit their pattern. However, the design process does not end there; in many cases they may have to create multiple punchcards and knit multiple textile samples before they are able to fabricate the punchcard that allow them to knit their desired result. Some options are available to allow a faster design process, such as an online punchcard generator that converts two color images to punchcard files that can be digitally cut [3], and another that uses text input to create similar SVG files [4]. Both options digitize the fabrication of punchcards but don't provide an interface that enables users to manipulate the design of the generated output. A different example uses OpenSCAD to create a code-based system that allows knitters to digitally design their punchcard using basic geometric shapes while visualizing the resulting punchcard [5]. However, this interface requires basic knowledge of code. In addition, all three generators allow for 2D patterning and do not explore the potential to use punchcards for shaping curved silhouettes and knitting 3D textures and shapes.

The third type of knitting machine is electronic. Early models used drawn Mylar sheets or manual digital input to program the machine. More modern electronic models are compatible with knitting software that allows the digital design process [1]. Although these machines provide easier design interface and fabrication, they are usually expensive and require purchasing design software. Therefore, many knitters prefer punch card knitting machines. New software and hacks created by makers encourage users to improve their old electronic domestic machines to allow computerized design and fabrication [6], [7], [8], [9], [10]. However, these improvements rely on the electronic parts that construct the machine and can't be applied to a punchcard-driven machine. We present a design tool that allows users to digitally design and fabricate punchcards for knitting 3D textures, long curved silhouettes, and 3D cone shapes on a punchcard knitting machine. Our tool relies on parametric design principles and allows knitters to manipulate their designs while automatically affecting the punchcard pattern. This process enables users to iterate before knitting, thus shortening the design process. In addition, our tool enables generating complex knitted structures, without having to rely on the expertise and knowledge of the user. The patterns are not limited to a standard-length size, rather, the punch cards may be fabricated in any length, and can be adjusted to work with any type of punch card knitting machine.

2 RELATED WORK

2.1 Digitizing manual processes

This work aims not only to expand the design opportunities of intricate 3D textures and shapes but also to provide a design interface to enable users to modify the desired result without having to knit the actual textile. Some workflows within the human-interaction world has been done to implement similar ideas in different fields. Digitizing the manual assembly process by providing a visualization interface that guides the user through creating a prototype composed of 3D parts and Lego bricks [11]. Another work allows users to reconstruct laser-cutting compatible files, enabling them to

customize the files and visualize their assembled structure before cutting them [12]. Other works focused on textile design and fabrication opportunities, exemplifying the automatization of otherwise manual processes in textile design. Such as enhancing a standard manual loom to enable componential design and semi-automatic fabrication of weaved textiles, [13] and creating an "un-dying" technique using programmable LEDs to explore self-de-staining textiles to achieve new aesthetic values and explore their uses as interactive materials [14].

2.2 Knitted structure research

While there has been growing interest in knitted structures, much of the work deals with design and fabrication using industrial knitting machines. Such work includes demonstrating the knitting of complex 3D shapes [15], and implementing AI to create a simpler image-based design process [16]. Some works directly indicate the connection between knitted structures across different fabrication methods. KnitGIST is a tool that allows users to influence distinctive features and generate knittable programs to knit on an industrial knitting machine. Still, it can also be used to generate instructions for hand knitting [17]. Another example is the work of Popescu et al which utilizes knitted structures as the base to form concrete architectural elements. In this case, the team used an electronic domestic knitting machine to fabricate a prototype concrete bridge before moving on to a concrete shell structure that was knitted on an industrial machine [18], [19].

2.3 Domestic knitting machines

Some works have focused on domestic knitting machines, enhancing the performances of an early electronic model to allow the computerized design of knitted pieces of clothing [20], and exploring the design of knittable tile patterns [21]. Taalman's and Yackel's work was motivated by punchcards but didn't specify a fabrication method or demonstrate a knitted result. In addition, their work focuses on 2D patterns and doesn't explore three-dimensional textures and shapes. Other studies in that field explore the idea of co-designing punchcards by machine and man. GAN models were used to generate new punchcards that were later traced off the computer screen and punched by hand [22]. Another work demonstrates a similar approach, using styleGAN2 to generate punchcard patterns and digitally process and fabricate them on a laser cutting machine [23]. Both studies utilize innovative technology that holds exciting potential applications, however as a design tool, they allow limited ability to manipulate the generated punchcard. Moreover, both focus on 2D patterning. In contrast, in our work, we focus on the fabrication of punchcards for 3D knitting of textures and shapes.

3 SYSTEM OVERVIEW

We present two workflows to design and fabricate computerized punchcards for 3D knitting and curved silhouette shaping. All swatches were knitted on a Passap Duomatic 80 (Figure 1), a punchcard driven double bed domestic knitting machine [24], [25]. In this machine, the needles are actuated by the Deco system, which is constructed of two devices: a reader and a selector. The first reads the card's pattern, and the other selects and moves pushers which



Figure 2: Standard punchcard (A). Knit/tuck pattern (B). Knit/no knit pattern (C) [29]

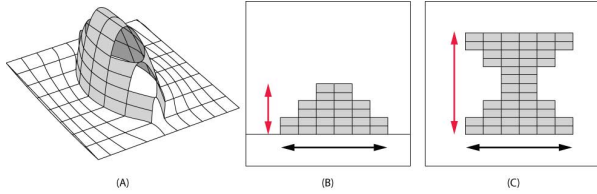


Figure 3: Spike module (A). Front view (B). Layout structure (C).

sets needles in work or hold positions. The reader can be set to advance the punchcard every 2 or 4 rows, allowing some design variations [24]. Since the knitted structure is affected by both the punchcard pattern and the knitting action set on the carriage, the same punchcard may be used to fabricate different knitted results. Figure 2 shows an example of such a case. In Figure 2 (B), the textile was knitted using knit and tuck stitches, in Figure 2 (C), the textile was knitted using only certain needles and skipping others, while also moving certain loops from the front to the back bed. In our workflow, we use the second method; knitting certain needles and skipping others. The knitter is only required to move the carriage from one side to the other in order to perform the actual knitting operation, without a need to intervene manually in the knitting process.

Our knitting workflows were created using Grasshopper for Rhinoceros. The punchcards were fabricated on Silhouette Cameo 4, a desktop cutting machine [26], [27], [28]. The Passap Deco reading mechanism allows a 40-needle width sequence. Since the reader is not mechanically limited to reading a specific length, we demonstrate different pattern lengths. Our workflow is compatible with Passap-typed punchcards but can be easily adjusted to be compatible with other punchcard proportions.

3.1 3D spike textures

Partial knitting (or short-row knitting) is a technique to knit 3D structures such as the heel of a sock. The 3D effect is accomplished by excessively knitting with only some of the operating needles, causing this area of the knitted structure to be longer than the rest. When a complete row is knitted again, the different areas of the textile join. The elongated segment curves and gives the textile its 3D shape [1], [30]. The first workflow is built on this logic. In figure 3, we show our 3D spike module (A), a front view (B), and a reconstructed layout of the 3D shape (C).

The width of the 3D module is set by the number of operating needles involved in its knitting; its height is directly affected by the number of rows that are knitted, which sets the length of the

module's punchcard layout accordingly. The automatic conversion process of the 3D visualization into a knittable punch card considers each module's location, size, and spacing. Since each punched hole of the card is being knitted twice to allow the carriage to move from right to left and back, the length of the entire pattern is recalculated accordingly. We use the center points of each visualized module as a reference to the location of their layout in the punchcard format (Figure 4 C). The maximal width of the modules was specifically designed to prevent overlapping and ensure the knitting of separate structures. The X value of the center points is calculated based on the relative location along the mesh. The Y value of this center point is recalculated to allow accurate spacing of different modules' sizes, considering the layout lengths of modules that appear before in the knitting sequence (Figure 4 D). After the center points are placed, the modules are reconstructed to generate the "hourglass" layout (Figure 4 D). Then the standard knitting rows that occupy the space between the modules are considered (Figure 4 E), resulting with an accurate punchcard pattern (Figure 4 F).

Our workflow allows users to manipulate different design features and visualize the knitted outcome before the punchcard and the knitted textile are fabricated. The workflow can be divided into four stages: (1) designing the visualization mesh, (2) automatically generating the punchcard, (3) digitally fabricating the punchcard, and (4) knitting the textile on a punchcard knitting machine.

First, users encounter our visualization of a knitted textile, its width is 40 needles long, and the user can determine its length by setting a number slider. Users can then introduce 3D spikes to the pattern, controlling the number of spike modules that appear on the X and Y axis of the mesh (Figure 5). We enable control over the diameter and height of the spikes by manipulating a number slider that controls all the spikes, or by using a greyscale image that automatically changes their size accordingly (Figure 6. A, B). To avoid overlapping and knitting problems, the spikes' minimal and maximal size values are relative to the number of modules that appear on the X-axis.

Users can also filter out spike modules based on a greyscale or black and white image. The number of spikes filtered can be adjusted using a number slider if necessary. This enhances the design opportunities beyond a rectangular grid of modules and allows the design of any other repeating outer shape.

Once the design of the visualization mesh is done, a DXF punchcard file is saved and cut using a Silhouette Cameo 4 desktop cutting machine. The punchcard is then fed into the knitting machine's reader and used to knit the desired result. In figure 6, we show knitted examples along with the corresponding images that generated the patterns and the punchcards used to knit them. In figures 6 A and B we show knitted textiles that were designed by using greyscale images. In Figure 6.C, we used a greyscale gradient to

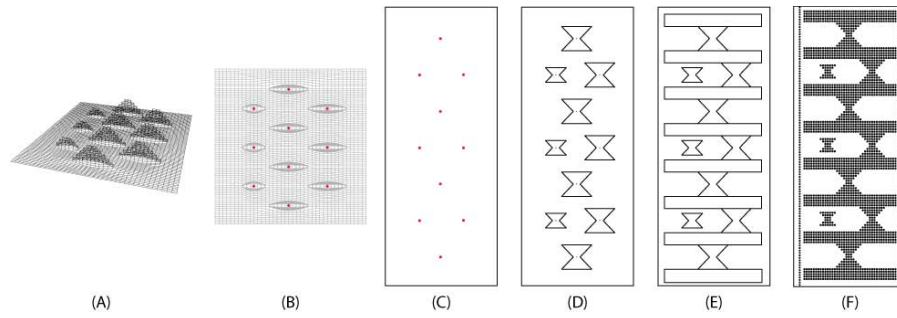


Figure 4: Visualization mesh(A). Top view of visualization mesh with modules' center points (B). layout center points placement (C) Layout construction (D). Normal knitting rows are considered (E). The resulted punchcard pattern (F).

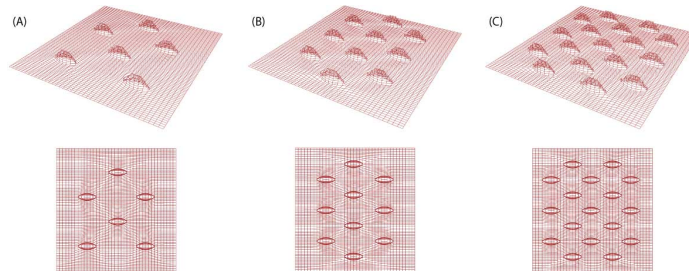


Figure 5: Module display variations represented as 3D meshes.

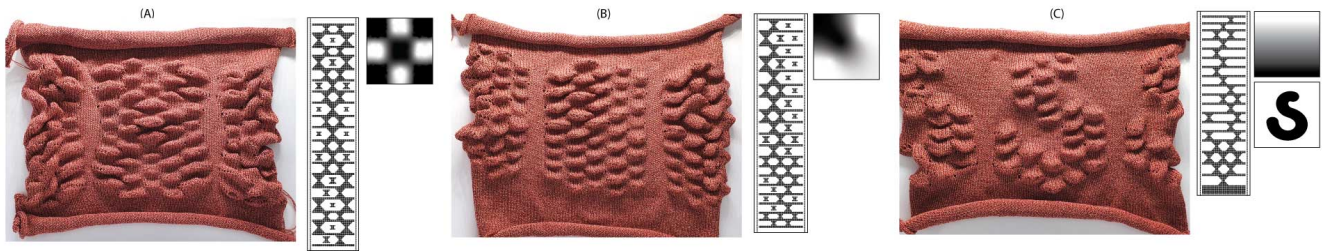


Figure 6: Examples of 3D textures, the corresponding images that generated the patterns and the punchcards that were knitted.

control the spike sizes and a black and white image of an S shape to filter some spikes and knit an S shape texture.

3.2 Curved surfaces and 3D cone shapes

In the second workflow, we utilize the machine's ability to knit curved surfaces for creating custom-made cured knits and 3D cone-based shapes. The same short-row principle allows us to knit curved flat surfaces by excessively repeating some operating needles. We demonstrate this idea by creating two design options. In the first option, users may create a curved surface based on a spline. In the second option, users may create 3D cone shapes constructed by long curved layouts. When using the spline variation, users may draw a curve on the Grasshopper-Rhino environment, which is then offset to a 40-needle width surface. The length of the surface is recalculated accordingly. This design workflow is currently compatible with curves without intersecting segments that allow offsetting 20 needle units in each direction. When using the cone

option, users first encounter a virtual 3D cone model. Users may manipulate its shape by moving number sliders which manipulate its top and bottom radii, since the cone proportions must be compatible with the punchcard, its height is recalculated according to its radii values. The cone shape is automatically converted into a corresponding curved layout.

In both design options, the mesh is automatically divided into smaller polygons (Figure 7 C), which maintain the maximal 40-needle width rule. A grid of rectangular cells parallel to the polygon segments is applied to the mesh, representing the knitting structure (Figure 7 D). The center line of each polygon is used as a guide to apply the polygon shapes to the punchcard. The spacing along the punchcard considers the distance between the center lines and the polygons' maximal height value, considering the polygons that appear before it in the knitting sequence.

We demonstrate the first design option using a U-shaped curve to generate a design, digitally cut a punchcard, and knit it on the domestic knitting machine (Figure 8). Most commercially available

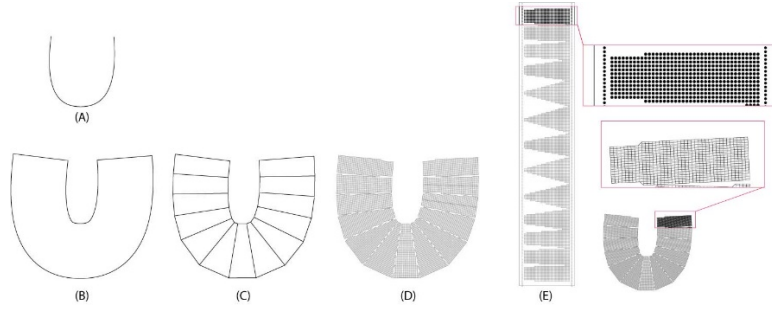


Figure 7: Spline is drawn in Rhino-Grasshopper environment (A). Corresponding offset (B). Polygon segments (C). Rectangular cell grid (D). Corresponding punchcard pattern (E).

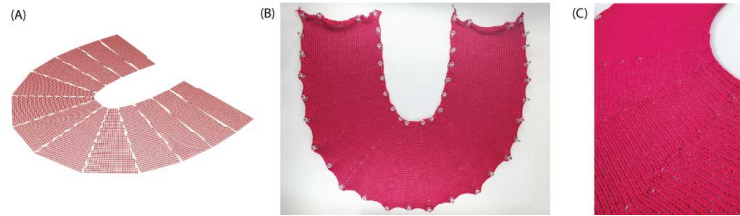


Figure 8: U-shaped visualization (A). U-shaped knitted textile (B). short row detail (C).

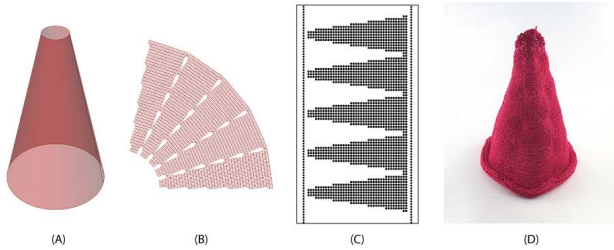


Figure 9: Digital 3D cone model(A). Visualization of its layout. Corresponding punchcard (C). Knitted cone (D).

punchcards allow for 2D patterning of rectangular textile shapes, in those cases shaping may be done by manipulating needles by hand. Short-row knitting is a common technique to achieve curved silhouettes whose overall width is larger than the number of needles being used in the knitting process. In the presented example, we illustrate one of the advantages of using our workflow, which utilizes the short-row knitting technique to knit long curved surfaces whose overall width exceeds the 40 needles restriction of the punchcard.

We demonstrate our second design option by creating a 3D cone shape, once the design process is done, a punchcard is digitally cut and used on the machine (Figure 9).

Since only the front needle bed of the Passap duo 80 knitting machine is actuated by the punchcard, our workflow usually involves knitting on only the front bed. For the 3D cone, however, the cast-on was done on both needle beds. The needles on the back bed were left in hold position while the needles on the front bed were knitted according to the punchcard pattern. This way, the yarn

remains connected to the back bed while the fabric elongates. After the punchcard pattern is knitted, the loops on the front bed are transferred to the back. The cast-off is done on the back needle bed, joining the start and the end edges together. Allowing the textile to achieve its final 3D shape without a need to process it post-knitting (Figure 10). This technique of creating seamless 3D cone structures relies on both needle beds. However, knitters of single-bed knitting machines may fabricate 3D cone shapes by using our workflow to knit their curved layouts and manually join their edges.

4 LIMITATIONS

Unlike modern electronic knitting machines which allow control over all front bed needles, our workflows rely on punchcards, therefore the generated patterns are restricted to a punchcard width (40 needles in the case of Passap duo 80 knitting machine). Since the punchcard is advanced every two rows, each short row in our workflow is being knitted twice. This causes small gaps in the shaped structure that are usually avoided when knitters manually intervene in the knitting operation.

5 DISCUSSION AND FUTURE WORK

Domestic knitting machines have the potential to fabricate intricate textile structures. The ability to realize this potential depends on the knitter's knowledge and skill, and in any case, requires long development that includes multiple trial and error stages. Our tool enables less experienced users to realize this potential by computerizing punchcards' design and fabrication process and allowing visualization. We digitize some of the manual processes, allowing knitters easier and faster design workflows when using punchcard-driven knitting machines.

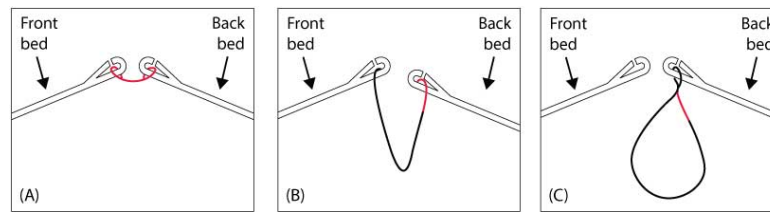


Figure 10: Cast-on was done on both beds (A). The punchcard was knitted only on the front bed, causing the textile to elongate (B). Loops were transferred from the front bed to the back bed and cast-off, allowing the structure to achieve its 3D shape (C).

First, we demonstrate 3D textures, utilizing the machines' ability to knit short-rows to create spike-like textures. This 3D spike pattern can be used for aesthetic purposes and is also suitable for knitting objects that require cushioning or would benefit from thickening the textile, such as socks, slippers, oven mittens, seats, and linings inside various pieces of clothing. Greyscale filtering may be used to enhance the personalization of the knitted object and may benefit both experienced and beginner knitters. Our second workflow utilized the short-rows knitting technique to knit long curved surfaces. Knitters may draw any curve that doesn't impose overlaps and allow a 40-needle offset to generate a punchcard accordingly. The punchcard reader is not limited to a certain card length and is, therefore, suitable for knitting long continuous patterns. We utilize this ability to knit cone layouts which are long curved textiles joined at the edges to form the 3D cone structure.

Future work includes expanding the variety of the knitted 3D modules, and further developing the customization of the knitted outcome to allow, for example, spikes within spikes, and 3D multi-color knitting. Additionally, we intend to explore the creation of long continuous curves to form larger-scale 3D shapes and intricate 3D cone-based shapes constructed by multiple conic units. Another interesting next step would be combining both techniques to allow textured shaping or complex 3D cone-based shapes, such as textured cones with star-like cross sections. Our workflow emphasizes how digital design and fabrication processes may bridge the gap between intricate parametric design and the ability to apply it to analog machines. Using our workflow, users can design complex structures on their computer, using parametric design tools, and fabricate these digital designs on a knitting machine without any electronic component.

REFERENCES

- [1] J. Sissons, *Knitwear: An Introduction to Contemporary Design*, 2nd ed. 2018. Accessed: Dec. 25, 2022. [Online]. Available: [https://web-p-ebscohost-com.ezlibrary.technion.ac.il/ehost/ebookviewer/ebook/ZTAwMHh3d19fMTcyODE0NF9fQU41?sid=\\$6ebf48b6-1560-4fc6-9429-ca746373175e@redis&vid\\$=\\$0&format\\$=\\$EB&rid\\$=\\$1](https://web-p-ebscohost-com.ezlibrary.technion.ac.il/ehost/ebookviewer/ebook/ZTAwMHh3d19fMTcyODE0NF9fQU41?sid=$6ebf48b6-1560-4fc6-9429-ca746373175e@redis&vid$=$0&format$=$EB&rid$=$1)
- [2] Instruction Manual for the Jones/Brother KH-588 Knitting Machine, BROTHER INDUSTRIES, LTD. Accessed: Jan. 09, 2023. [Online]. Available: https://mkmanuals.com/downloadable/download/sample/sample_id/161/
- [3] "Knitme." <https://oknitme.serpicanaro.org/#> (accessed Jan. 08, 2023).
- [4] "Punchcard Generator - Brenda A. Bell." <https://brendaabel.com/knittingtools/pgenerator/> (accessed Jan. 08, 2023).
- [5] "Punch Card Knitting Machine Patterns with OpenSCAD - mathgrrl." <https://mathgrrl.com/hacktastic/2019/05/punch-card-knitting-machine-patterns-with-openscad/> (accessed Jan. 08, 2023).
- [6] "Hacking the Brother KH-930e knitting machine - Becky Stern." <https://beckystern.com/2010/11/02/hacking-the-brother-kh-930e-knitting-machine/> (accessed Jan. 13, 2023).
- [7] "AYAB - all yarns are beautiful." <https://ayab-knitting.com/> (accessed Jan. 13, 2023).
- [8] "img2track." <https://daviworks.com/knitting/> (accessed Dec. 25, 2022).
- [9] "The Knitting Network Printer - Electromagnetic Field." <https://www.emfcamp.org/schedule/2018/29-the-knitting-network-printer> (accessed Dec. 25, 2022).
- [10] "multi threaded banjo dinosaur knitting adventure 2D extreme!!! 8-bit hacked knitting machine game." <https://www.mediamatic.net/en/page/14004/multi-threaded-banjo-dinosaur-knitting-adventure-2d-extreme> (accessed Dec. 25, 2022).
- [11] S. Mueller, T. Mohr, K. Guenther, J. Frohnhofen, and P. Baudisch, "FaBrickation: Fast 3D printing of functional objects by integrating construction kit building blocks," in *Conference on Human Factors in Computing Systems - Proceedings*, 2014, pp. 3827–3834. doi: 10.1145/2556288.2557005.
- [12] T. Roumen *et al.*, "Assembler3: 3D Reconstruction of Laser-Cut Models," May 2021, pp. 1–11. doi: 10.1145/3411764.3445453.
- [13] L. Albaugh, J. McCann, L. Yao, and S. Hudson, "Enabling personal computational handweaving with a low-cost jacquard loom," in *Conference on Human Factors in Computing Systems - Proceedings*, May 2021. doi: 10.1145/3411764.3445750.
- [14] F. Bell *et al.*, "Self-detecting textiles: Designing interactive systems with fabric, stains and light," in *Conference on Human Factors in Computing Systems - Proceedings*, May 2021. doi: 10.1145/3411764.3445155.
- [15] V. Narayanan, L. Albaugh, J. Hodgins, S. Coros, and J. McCann, "Automatic machine knitting of 3d meshes," *ACM Trans Graph*, vol. 37, no. 3, Jul. 2018, doi: 10.1145/3186265.
- [16] A. Kaspar, T.-H. Oh, L. Makatura, P. Kellnhofer, J. Aslars, and W. Matusik, "Neural Inverse Knitting: From Images to Manufacturing Instructions," Feb. 2019, [Online]. Available: <http://arxiv.org/abs/1902.02752>
- [17] M. Hofmann, J. Mankoff, and S. E. Hudson, "KnitGIST: A programming synthesis toolkit for generating functional machine-knitting textures," in *UIST 2020 - Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*, Oct. 2020, pp. 1234–1247. doi: 10.1145/3379337.3415590.
- [18] M. Popescu, L. Reiter, A. Liew, T. van Mele, R. J. Flatt, and P. Block, "Building in Concrete with an Ultra-lightweight Knitted Stay-in-place Formwork: Prototype of a Concrete Shell Bridge," *Structures*, vol. 14, pp. 322–332, Jun. 2018, doi: 10.1016/j.istruc.2018.03.001.
- [19] M. Popescu *et al.*, "Structural design, digital fabrication and construction of the cable-net and knitted formwork of the KnitCandela concrete shell," *Structures*, vol. 31, pp. 1287–1299, Jun. 2021, doi: 10.1016/j.istruc.2020.02.013.
- [20] C. Do-Seung, K.-B. Kim, and J.-S. Jang, "A Study on 3D Printer Design for Clothing Printing: Focusing on Knitted Wearable Clothing Output," *International Journal of Advanced Smart Convergence*, vol. 9, no. 1, pp. 82–89, 2020, doi: 10.7236/IJASC.2020.9.1.82.
- [21] L. Taalman and C. Yackel, "Wallpaper Patterns for Lattice Designs".
- [22] K. Geck, "Knitting Algorithmic Assemblages," *TEXTILE*, pp. 1–18, Nov. 2022, doi: 10.1080/14759756.2022.2114240.
- [23] V. Melynyk, "Punch Card Knitting Pattern Design in Collaboration with GAN," 2021.
- [24] deco, MADAG Maschine-und Apparatebau Dietikon AG. Dietikon, Switzerland.
- [25] PASSAP duomatic 80, MADAG Maschine-und Apparatebau Dietikon AG. MADAG Maschine-und Apparatebau Dietikon AG, Dietikon, Switzerland.
- [26] "Grasshopper - algorithmic modeling for Rhino." <https://www.grasshopper3d.com/> (accessed Jan. 17, 2023).
- [27] "Rhino - Rhinoceros 3D." <https://www.rhino3d.com/> (accessed Jan. 17, 2023).
- [28] "Silhouette America - Silhouette Cameo®." <https://www.silhouetteamerica.com/featured-product/cameo> (accessed Jan. 17, 2023).
- [29] deco 101-330, MADAG Maschine-und Apparatebau Dietikon AG. Dietikon, Switzerland.
- [30] Alessandrina, "Holding stitches/ short rows - alessandrina.com," 2013. <https://alessandrina.com/2013/12/18/holding-stitches-short-rows/> (accessed Jan. 14, 2023).